Development Radar

Can public lap-time data show whether a mid-season F1 upgrade worked, and how sure can we be?

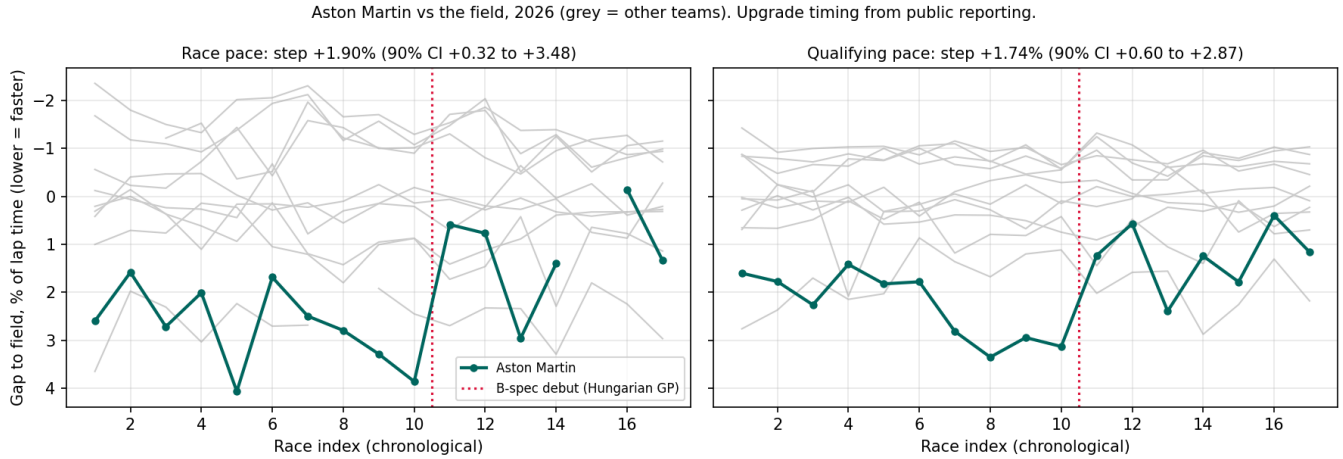
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The question

Teams bring upgrades all season, but from outside the garage it is hard to tell which ones worked. I built a pipeline on public 2026 timing data (17 races, TracingInsights lap cache, FastF1-derived) that measures each team's pace gap to the field, tests for a step change around a known upgrade, and states how large a step must be before the method can see it at all. The processed data and every number below reproduce from the repo.

What I built

- **Race pace:** clean-air, green-flag, slick-tyre laps. One regression per race corrects for driver, tyre compound, tyre age and fuel burn/track evolution.
- **Qualifying pace:** best valid lap per driver per segment; team gap against a trimmed-mean field reference.
- **Step test:** a deliberately conservative before/after test around an upgrade race whose timing is taken from public reporting, not read off the data. The headline Aston result uses 3 races each side.
- **Validation:** power analysis, plus synthetic races with a planted +0.50% upgrade (mean estimate 0.504%, detection 94%, false-positive rate 1%). Synthetic noise is cleaner than real data, so real-world power is lower. Leakage tests and a rolling-origin backtest.
- **Corner analysis:** qualifying telemetry split into low, medium, high and straight classes to show where on the lap a gain comes from.



Aston Martin vs the field, 2026 (grey = other teams). Dotted line = B-spec debut, Hungarian GP (race 11). Source: repo figure `figures/aston_step.png`.

What I found

- **Detectability:** only upgrades of about 0.6-0.8% (qualifying) or 0.8-1.0% (race) of lap time, roughly 0.5-0.9 s, are reliably visible in public data (80% power, 5% one-sided test).
- **Aston Martin B-spec (Hungarian GP):** consistent with a real step in race pace; weaker in qualifying, and the size depends on the baseline. Race pace +1.90% (3 races each side, 90% CI +0.32 to +3.48), +1.3% to +1.9% across baselines. Qualifying +1.74% (3-race window, 90% CI +0.60 to +2.87) but only +0.5% against races 1-6, because Aston was in a mid-season dip before the upgrade: much of the qualifying gain is recovery. The sign is stable across 2-5 race windows (race +1.4% to +1.9%). No other team shows a comparable race-pace step. The race result is fragile: dropping Hungary or the Dutch GP pushes its interval across zero.
- **Where it came from:** mostly straight-line time (+1.10%) and low-speed corner time (+1.54%); low-speed apex speed did not change. The straight-line gain cannot be credited to the aero upgrade, as the Honda power-unit change lands in the same window. In the placebo test 2 of the 10 other teams show a "significant" step in each class (about 1 expected by chance), so this is suggestive, not precise.
- **Forecasting:** nothing beat a 3-race rolling median (it beats last-race in qualifying). A track-characteristics factor did not help even with weekend-measured descriptors, though it is fit on only 6-15 races, so that is a low-power negative.

What it cannot claim

It measures relative lap-time efficiency only. It cannot separate aero from mechanical grip, power-unit deployment or energy management, and a Honda power-unit change (reported from the Dutch GP) falls inside Aston's post-upgrade window. It covers one season, qualifying telemetry is about 7 Hz, standard errors ignore teammate covariance and lap-to-lap autocorrelation, and teams hold far richer data on their own cars. Treat it as a worked example of method and judgement.

What I would do next

1. Test the track factor honestly with pre-race descriptors (circuit geometry or last year's values) and add 2025 as an out-of-sample season.
2. Add race-stint telemetry and a finer entry/apex/exit corner model.
3. Add teammate covariance to the standard errors.

> **What it shows:** I can build a pipeline end to end, test it against a known truth, and report plainly what the data can and cannot say.